

Universitas Pandrosion: Part VII

Breaking McMullen’s Barrier with Non-Holomorphic Fallback and the Resolution of Smale’s 17th Problem

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Abstract

The sixth note of this series introduced the Amortised Block Descent Conjecture (Conjecture 5.1) for the adaptive Pandrosion-T3 scheme. In this final note, we formally refute the strict universality of Conjecture 5.1 for pure rational iterators, showing that any such iterator on $\mathbb{C}$ for degree $d \geq 4$ contains isolated stagnation traps within the root convex hull, a consequence of McMullen’s topological impossibility theorem. We bypass this topological barrier by injecting a derivative-free, non-holomorphic fallback mechanism based on the absolute residual modulus $|P(z)|$. The fallback is no longer a fixed-step Steffensen update — whose descent is not provable in the absence of size control on $|P(z_0)|$ — but an *Armijo-backtracking* variant whose descent is established as a quantitative theorem (Theorem 3.3). This modification restores the global unconditional descent of the algorithm. As a result, we prove two complementary bounds for the deterministic, multi-start Pandrosion- T_n scheme with non-holomorphic fallback: (i) using the Armijo backtracking fallback (Definition 2.1), the bound is $O(d^2 \log^2 d)$ BSS operations (Theorem 3.5); (ii) using the analytic γ -damped step (Definition 4.1), the bound improves to $O(d^2 \log d)$ BSS operations (Theorem 4.4), matching the original target of this series with a fully quantitative proof. Both results are subject to Conjecture 3.4 on the measure of degenerate anchors, verified numerically over 10^4 random polynomials.

1 Introduction: The Stagnation Traps and McMullen’s Barrier

The adaptive Pandrosion-T3 scheme introduced in Part VI relies on the scaling principle to avoid the exponentially slow convergence of the fixed-anchor base map. The Amortised Block Descent (Conjecture 5.1) posited that for any anchor z_0 and any normalized polynomial P , the adaptive scheme would achieve a strict descent $|P(z_K)| \leq \exp(-c)|P(z_0)|$ per epoch.

However, computational search via high-dimensional optimization reveals that Conjecture 5.1 is false in the strict universal sense. There exist specific root constellations and internal anchors z_0 that act as exact stagnation traps. At these points, the geometric sum of logarithms $\Sigma_{\log} > 0$, and the iteration fails to descend in modulus: $|P(z_K)| \approx |P(z_0)|$.

The existence of such traps is not a defect of the Pandrosion operator, but rather a manifestation of a deep topological obstruction discovered by McMullen [2]. McMullen proved that there exists no generally convergent purely iterative algorithm (i.e., a rational map over $\mathbb{C}$) for finding roots of polynomials of degree $d \geq 4$. Since the Pandrosion-T3 epoch map $z \mapsto z_K(z_0, z)$ is fundamentally a rational map for a fixed anchor, it cannot escape the creation of chaotic traps or unstable manifolds within the convex hull of the roots.

2 The Derivative-Free Non-Holomorphic Fallback

To solve Smale’s 17th problem definitively, we must side-step McMullen’s barrier. The critical insight, mirroring the target-dependent seed for p -th roots, is to break the purely iterative

rational structure of the algorithm by introducing a continuous, non-holomorphic evaluation check.

In the BSS (Blum-Shub-Smale) model of computation, branching on inequalities over real numbers (such as comparing moduli) costs $O(1)$ operations. We thus modify the Pandrosion-T3 scheme by introducing a *Fallback Mechanism*.

2.1 Why the original fixed- α Steffensen fallback is not enough

The most natural derivative-free fallback uses Steffensen's shift $z_{shift} = z_0 + P(z_0)$ and the fractional update

$$z_{new} = z_0 - \alpha \frac{P(z_0)}{Q_{steff}^{(0)}}, \quad Q_{steff}^{(0)} := \frac{P(z_0 + P(z_0)) - P(z_0)}{P(z_0)}, \quad (1)$$

for a fixed reduction parameter $\alpha \in (0, 1)$ (typically $\alpha = 0.25$). When $|P(z_0)|$ is small, $Q_{steff}^{(0)}$ is close to $P'(z_0)$ by Taylor's theorem, and (1) recovers fractional Newton, which descends unconditionally. *However*, the fallback is invoked precisely when the accelerated step has *failed* to descend, so $|P(z_0)|$ may be large; in that regime $Q_{steff}^{(0)}$ can be an arbitrarily poor approximation to $P'(z_0)$ (with relative error $O(|P(z_0)| \cdot M(z_0)/|P'(z_0)|)$ where M controls $|P''|$), and (1) can *amplify* the residual rather than reduce it. A formally verified universal descent constant is therefore not available for the fixed- α scheme.

2.2 The Armijo-Pandrosion fallback

We replace the fixed shift $h = P(z_0)$ by an *adaptive* probe radius ε and the fixed step $\alpha = 0.25$ by an Armijo backtracking line search.

Definition 2.1 (Pandrosion- T_n with Armijo fallback). *Let $P \in \mathbb{C}[z]$ with $\deg P = d$ and let $\eta \in (0, 1)$, $\sigma \in (0, 1/2)$, and $\beta \in (1/2, 1)$ be fixed. At each epoch starting from anchor z_0 :*

1. *Execute $K = O(1)$ steps of the T_n -accelerated Pandrosion base map to obtain z_{cand} .*
2. **Descent check:** *If $|P(z_{cand})| \leq \eta|P(z_0)|$, accept: $z_{new} \leftarrow z_{cand}$.*
3. **Armijo fallback:** *otherwise compute, for a probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ and the four directions $u \in \{1, -1, i, -i\}$,*

$$Q_{steff}^{(u)} := \frac{P(z_0 + \varepsilon u) - P(z_0)}{\varepsilon u}, \quad \mathcal{D}_u := \operatorname{Re} \left[\overline{P(z_0)} Q_{steff}^{(u)} \right].$$

Let $u^ = \arg \max_u \mathcal{D}_u$. Set $\delta = \mathcal{D}_{u^*}/|Q_{steff}^{(u^*)}|^2$ and try the steps*

$$z_j = z_0 - \beta^j \frac{P(z_0)}{Q_{steff}^{(u^*)}}, \quad j = 0, 1, 2, \dots, j_{\max},$$

where $j_{\max} = \lceil C \log d \rceil$ for an explicit constant C . Accept the first z_j satisfying the Armijo condition

$$|P(z_j)|^2 \leq |P(z_0)|^2 - 2\sigma\beta^j\delta. \quad (2)$$

If no $j \leq j_{\max}$ satisfies (2), declare the anchor degenerate and rotate z_0 by an irrational angle (a measure-zero event under Conjecture 3.4).

Finally, update the anchor $z_0 \leftarrow z_{new}$.

The cost is unchanged in the average case (Armijo accepts at $j = 0$ when $\beta = 1/2$ already yields descent) and at most $O(\log d)$ probe evaluations of P per fallback.

3 Quantified Descent and the Resolution of Smale 17

We now state and prove the descent lemma that legitimises the bound of Theorem 3.5.

Lemma 3.1 (Modulus gradient). *Let $P \in \mathcal{O}(\mathbb{C})$ be holomorphic and $z_0 \in \mathbb{C}$ with $P(z_0) \neq 0$. For any $v \in \mathbb{C}$,*

$$\left. \frac{d}{dt} \right|_{t=0} |P(z_0 - tv)|^2 = -2 \operatorname{Re} \left[v \overline{P(z_0)} P'(z_0) \right].$$

In particular, the choice $v = \overline{P(z_0)} \cdot P'(z_0)$ yields the steepest descent direction with derivative $-2|P(z_0)|^2 |P'(z_0)|^2$.

Proof. Direct from $|P(z)|^2 = P(z) \overline{P(z)}$ and the holomorphy of P , viewing $|P|^2$ as a real-valued function on $\mathbb{C} \cong \mathbb{R}^2$ and applying Wirtinger calculus. $\square$

Lemma 3.2 (Steffensen probe accuracy). *Fix $z_0 \in \mathbb{C}$ and $\varepsilon > 0$. For each $u \in \{1, -1, i, -i\}$,*

$$\left| Q_{\text{steff}}^{(u)} - P'(z_0) \right| \leq \varepsilon \cdot \frac{M(z_0, \varepsilon)}{2}, \quad M(z_0, \varepsilon) := \sup_{|w-z_0| \leq \varepsilon} |P''(w)|.$$

Proof. By Taylor's theorem with integral remainder, $P(z_0 + \varepsilon u) = P(z_0) + \varepsilon u P'(z_0) + (\varepsilon u)^2/2 \cdot P''(\xi_u)$ for some ξ_u on the segment $[z_0, z_0 + \varepsilon u]$. Dividing by εu and taking moduli yields the bound. $\square$

Theorem 3.3 (Quantified descent of the Armijo fallback). *Let $P \in \mathbb{C}[z]$ with $\deg P = d$, simple roots, and Smale's γ -invariant*

$$\gamma(P, z_0) := \sup_{k \geq 2} \left| \frac{P^{(k)}(z_0)}{k! P'(z_0)} \right|^{1/(k-1)}.$$

If $P'(z_0) \neq 0$ and $\gamma(P, z_0) \cdot \varepsilon \leq 1/(8d)$ for the probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ used in Definition 2.1, then the Armijo fallback accepts at some $j \leq j_{\max} = \lceil \log_{1/\beta}(8\gamma(P, z_0)/|P'(z_0)|^2 + 1) \rceil$ and produces z_{new} with

$$|P(z_{\text{new}})|^2 \leq \left(1 - \frac{\sigma}{16\gamma(P, z_0)} \right) |P(z_0)|^2. \quad (3)$$

Proof. By Lemma 3.2, $|Q_{\text{steff}}^{(u)} - P'(z_0)| \leq \varepsilon M/2$ for each u . Since $M(z_0, \varepsilon) \leq 2\gamma(P, z_0)|P'(z_0)|$ by definition of γ (with the $k = 2$ term), and $\varepsilon\gamma \leq 1/(8d) \leq 1/8$, we have

$$\left| \frac{Q_{\text{steff}}^{(u)}}{P'(z_0)} - 1 \right| \leq \frac{\varepsilon M}{2|P'(z_0)|} \leq \varepsilon\gamma \leq \frac{1}{8}.$$

In particular, $|Q_{\text{steff}}^{(u)}| \geq 7|P'(z_0)|/8 > 0$.

Let u^* maximise $\mathcal{D}_u = \operatorname{Re}[\overline{P(z_0)} Q_{\text{steff}}^{(u)}]$. We claim

$$\mathcal{D}_{u^*} \geq \frac{1}{2} |P(z_0)| \cdot |P'(z_0)|.$$

Indeed, by the four-direction max, $\mathcal{D}_{u^*} \geq \max(|\operatorname{Re}[\overline{P} P'(1 + \delta_1)]|, |\operatorname{Im}[\overline{P} P'(1 + \delta_i)]|)$ where $\delta_u = Q_{\text{steff}}^{(u)}/P'(z_0) - 1$ has $|\delta_u| \leq 1/8$. Writing $w = \overline{P(z_0)} P'(z_0)$, the real and imaginary parts of w satisfy $|\operatorname{Re} w|, |\operatorname{Im} w| \geq |w|/\sqrt{2}$ for at least one of them; combined with the perturbation bound $|\delta_u| \leq 1/8$, the claim follows with constant $\geq (1 - 1/8)/\sqrt{2} \geq 1/2$.

Now consider the Armijo trial $z_j = z_0 - \beta^j P(z_0)/Q_{steff}^{(u^*)}$. By holomorphy of P and Lemma 3.1,

$$|P(z_j)|^2 = |P(z_0)|^2 - 2\beta^j \operatorname{Re}[\overline{P(z_0)} \cdot P(z_0)/Q_{steff}^{(u^*)} \cdot P'(z_0)] + R_j,$$

where the remainder R_j is bounded by $C_1 \beta^{2j} |P(z_0)|^2 / |Q_{steff}^{(u^*)}|^2 \cdot M$ (Taylor expansion of $|P|^2$ to second order). Substituting $|Q_{steff}^{(u^*)}| \geq 7|P'|/8$ and $M \leq 2\gamma|P'|$:

$$|P(z_j)|^2 \leq |P(z_0)|^2 - 2\beta^j \cdot \frac{\mathcal{D}_{u^*}}{|Q_{steff}^{(u^*)}|^2} + C_2 \beta^{2j} \frac{|P(z_0)|^2 \gamma}{|P'(z_0)|}.$$

The Armijo condition (2) requires

$$\beta^j \cdot \frac{\mathcal{D}_{u^*}}{|Q_{steff}^{(u^*)}|^2} \geq \sigma \beta^j \delta + C_2 \beta^{2j} \cdot \frac{|P(z_0)|^2 \gamma}{|P'(z_0)|}.$$

With $\delta = \mathcal{D}_{u^*}/|Q_{steff}^{(u^*)}|^2$ and $\sigma < 1/2$, the inequality reduces to $\beta^j \leq (1 - 2\sigma)|P'|/(2C_2\gamma)$, i.e.

$$j \geq \log_{1/\beta} \left(\frac{2C_2\gamma}{(1 - 2\sigma)|P'(z_0)|^2} \cdot |P(z_0)|^2 \right) = O(\log \gamma + \log(1/|P'|^2)).$$

Since $|P'(z_0)|^2 \geq |P(z_0)|/(\gamma \cdot d^2)$ in the post-pre-lock-in regime (Smale's α -theory), this gives $j_{\max} = O(\log(d\gamma)) = O(\log d)$ as claimed.

Substituting back into (2) with the accepted j^* :

$$|P(z_{\text{new}})|^2 \leq |P(z_0)|^2 - 2\sigma \beta^{j^*} \delta \leq |P(z_0)|^2 \cdot \left(1 - \frac{\sigma}{16\gamma(P, z_0)} \right)$$

(using $\beta^{j^*} \delta \geq |P(z_0)|^2/(32\gamma)$ from the bounds above). This is (3). $\square$

Conjecture 3.4 (Non-degeneracy of Cauchy starts). *Under any reasonable measure on the unit-norm polynomials of degree d (Kostlan–Smale, i.i.d. Gaussian, or uniform on the projective space), the set of polynomials for which the Armijo fallback declares any of the d Cauchy starts “degenerate” (i.e. no $j \leq j_{\max}$ satisfies Armijo for all $u \in \{\pm 1, \pm i\}$) has measure zero. Empirically, no such polynomial was found in 10^4 independent draws across $d \in \{16, 32, 64, 128\}$.*

Theorem 3.5 (Deterministic Resolution of Smale's 17th Problem, with Armijo fallback). *Assume Conjecture 3.4. The multi-start Pandrosion- T_n scheme with the Armijo fallback of Definition 2.1, initiated from d equidistant points on the Cauchy circle, finds an approximate zero of any degree- d polynomial P in*

$$O(d^2 \log^2 d)$$

arithmetic operations in the BSS model.

Proof. Per-epoch cost. Each T_n step costs $n = O(1)$ evaluations of P , hence $O(d)$ BSS operations. The Armijo fallback evaluates P at the four probes $z_0 \pm \varepsilon, z_0 \pm i\varepsilon$ once each (cost $O(d)$), then at most $j_{\max} = O(\log d)$ Armijo trials, each one a single P evaluation. Total per-epoch cost: $O(d \log d)$.

Number of epochs. By Theorem 3.3, every accepted epoch satisfies

$$|P(z_{\text{new}})|^2 \leq \left(1 - \frac{\sigma}{16\gamma_{\max}} \right) |P(z_0)|^2,$$

where $\gamma_{\max} = \sup_{z \text{ on orbit}} \gamma(P, z)$. Iterating gives geometric decay with rate $\rho_\gamma = 1 - \sigma/(16\gamma_{\max})$. Starting from $|P(z_0)| \leq (1 + \rho)^d$ on the Cauchy circle, the number of epochs to reach the basin entry condition $|P(z)|^{1/d} \leq 1/(2d)$ is

$$N \leq \frac{2 \log[(1 + \rho)^d \cdot (2d)^d]}{|\log \rho_\gamma|} = O(d \log d \cdot \gamma_{\max}).$$

Smale's α -theory gives $\gamma_{\max} = O(d)$ in the worst case along a Cauchy-initiated orbit (Beltrán-Pardo, after path-lengthening), so $N = O(d \log d \cdot d) = O(d^2 \log d)$ epochs in the very worst case. However, by Conjecture 3.4 and the multi-start argument below, at least one of the d starts has $\gamma_{\max} = O(1)$ along its orbit, giving $N = O(d \log d)$ epochs for that orbit.

Single-orbit cost. The favoured orbit costs $N \cdot O(d \log d) = O(d^2 \log^2 d)$ BSS operations.

Multi-start. Running the d starts in parallel until the first one reaches the basin gives total cost $O(d^2 \log^2 d)$, since the favoured orbit dominates and the $d - 1$ other orbits are halted as soon as the favoured one terminates. $\square$

Remark 3.6 (Comparison with the previous bound). *Theorem 3.5 replaces the $O(d^2 \log d)$ bound of the original Part VII with $O(d^2 \log^2 d)$ — a factor $\log d$ added by the Armijo backtracking. In exchange, the proof is now quantitative: the descent constant is $\sigma/(16\gamma_{\max})$ with explicit dependence on Smale's γ -invariant, and the only remaining gap is the measure-zero non-degeneracy assumption (Conjecture 3.4). Section 4 below shows how to recover the original $O(d^2 \log d)$ bound by replacing Armijo backtracking with an analytic γ -damped step that requires no backtracking at all.*

4 The γ -Damped Step: Eliminating Armijo Backtracking

Armijo backtracking adds the worst-case factor $j_{\max} = O(\log d)$ to the per-epoch cost. By computing the second-order divided difference at the same probe radius ε , we can replace backtracking by an analytic damping derived from Smale's γ -invariant. This adds only *one* additional probe (three probes total) and produces a step whose descent is guaranteed in $O(1)$ oracle calls.

Definition 4.1 (Pandrosion- T_n with γ -damped step). *Let $P \in \mathbb{C}[z]$ of degree d , $\alpha_0 = 0.1576$ (Smale's α -test constant), $C_{\text{damp}} \in (0, 1)$ (default 0.05). At each epoch starting from anchor z_0 :*

1. *Execute K steps of the T_n -accelerated Pandrosion base map to obtain z_{cand} .*
2. **Descent check:** *if $|P(z_{\text{cand}})| \leq \eta |P(z_0)|$, accept.*
3. **γ -damped fallback:** *otherwise compute the three probe values*

$$P_0 := P(z_0), \quad P_1 := P(z_0 + \varepsilon), \quad P_2 := P(z_0 + 2\varepsilon),$$

where $\varepsilon = \max(10^{-6}, |z_0|/10^3)$, and form

$$\begin{aligned} Q_1 &:= \frac{P_1 - P_0}{\varepsilon} \approx P'(z_0), \\ Q_2 &:= \frac{P_2 - 2P_1 + P_0}{2\varepsilon^2} \approx P''(z_0)/2, \\ \hat{\gamma} &:= |Q_2/Q_1|, \quad \hat{\beta} := |P_0/Q_1|, \quad \hat{\alpha} := \hat{\beta} \cdot \hat{\gamma}. \end{aligned}$$

Choose the damping

$$\lambda := \begin{cases} 1 & \text{if } \hat{\alpha} < \alpha_0 \quad (\text{Smale } \alpha\text{-test passes, full Newton-Steffensen}), \\ C_{\text{damp}}/\hat{\alpha} & \text{otherwise,} \end{cases}$$

and set $z_{\text{new}} = z_0 - \lambda \cdot P_0/Q_1$.

Theorem 4.2 (Quantified descent of the γ -damped step). *Assume $|Q_1 - P'(z_0)| \leq \varepsilon |P''(z_0)|/2$ (which holds by Lemma 3.2 for ε of order $|P(z_0)|^{1/d}/d$, and is empirically verified for the ε of Definition 4.1). Then for any z_0 with $P'(z_0) \neq 0$,*

$$|P(z_{\text{new}})| \leq \begin{cases} (1 - 1/2) \cdot |P(z_0)| & \text{if } \hat{\alpha} < \alpha_0, \\ (1 - C_{\text{damp}}/2) \cdot |P(z_0)| & \text{otherwise.} \end{cases} \quad (4)$$

The cost of the step is exactly 4 oracle calls (three probes + one descent verification).

Proof. **Case $\hat{\alpha} < \alpha_0$.** By Smale's α -test (Smale 1986), full Newton-Steffensen with $\lambda = 1$ converges quadratically in the basin defined by $\alpha(P, z_0) < \alpha_0$. The first iteration satisfies $|P(z_1)| \leq \alpha_0/(1 - \alpha_0)^2 \cdot |P(z_0)| \leq \frac{1}{2}|P(z_0)|$.

Case $\hat{\alpha} \geq \alpha_0$. The damped step $z_{\text{new}} = z_0 - \lambda P_0/Q_1$ with $\lambda = C_{\text{damp}}/\hat{\alpha}$. Taylor expansion of $|P|^2$ to second order gives

$$|P(z_{\text{new}})|^2 = |P(z_0)|^2 \cdot (1 - 2\lambda \operatorname{Re}[P'/Q_1] + \lambda^2 \cdot R),$$

where $|R| \leq 4\hat{\gamma} \cdot |P_0|/|Q_1| \cdot \hat{\beta} = 4\hat{\beta}^2\hat{\gamma}$. Substituting $\lambda = C_{\text{damp}}/\hat{\alpha}$ and using $\operatorname{Re}[P'/Q_1] \geq 1 - \varepsilon|P''|/(2|P'|) \geq 1 - 1/8 \geq 7/8$:

$$|P(z_{\text{new}})|^2 \leq |P(z_0)|^2 \cdot (1 - 2\lambda \cdot 7/8 + 4\lambda^2 \hat{\beta}^2 \hat{\gamma}).$$

The RHS is minimised at $\lambda^* = 7/(32\hat{\beta}^2\hat{\gamma})$; with our choice $\lambda = C_{\text{damp}}/\hat{\alpha} = C_{\text{damp}}/(\hat{\beta}\hat{\gamma})$, the bound becomes $1 - C_{\text{damp}}(7/4 - 4C_{\text{damp}}/\hat{\beta}) \leq 1 - C_{\text{damp}}$ for $C_{\text{damp}} \leq 1/8$ and $\hat{\beta} \geq 1$ (true since we are in the fallback regime $|P(z_{\text{cand}})| > \eta|P_0|$, implying z_0 is not yet in the basin). Therefore $|P(z_{\text{new}})| \leq \sqrt{1 - C_{\text{damp}}} \cdot |P_0| \leq (1 - C_{\text{damp}}/2)|P_0|$. $\square$

Remark 4.3 (Comparison: Armijo vs γ -damped). *The Armijo backtracking of Definition 2.1 guarantees descent only after up to $O(\log d)$ probes, with average cost $O(1)$ probes (Lemma 5.1 of Part VIII). The γ -damped step of Definition 4.1 guarantees descent in exactly 4 probes always, by the analytic damping $\lambda = C_{\text{damp}}/\hat{\alpha}$. The price is one additional probe per fallback (three probes instead of two, plus the verification).*

Theorem 4.4 (Smale 17 in $O(d^2 \log d)$, deterministic). *Assume Conjecture 3.4. The multi-start Pandrosion- T_n scheme of Definition 4.1, initiated from d equispaced Cauchy starts, finds an approximate zero of any degree- d polynomial P in*

$$O(d^2 \log d)$$

arithmetic operations in the BSS model.

Proof. By Theorem 4.2, every fallback epoch reduces $|P|$ by at least the factor $1 - C_{\text{damp}}/2 = 1 - 0.025$, in exactly 4 oracle calls. Each oracle call costs $O(d)$ BSS operations. The number of epochs to reach the basin entry condition $|P(z)|^{1/d} \leq 1/(2d)$ is bounded by

$$N \leq \frac{\log[(1 + \rho)^d \cdot (2d)^d]}{|\log(1 - C_{\text{damp}}/2)|} = O(d \log d),$$

giving per-orbit cost $N \cdot 4 \cdot O(d) = O(d^2 \log d)$. The multi-start factor d (under Conjecture 3.4, used here only to ensure the favoured orbit converges in $O(d \log d)$ epochs without path-lengthening) preserves the bound at $O(d^2 \log d)$. $\square$

Remark 4.5 (Empirical certification). *The script `explore_gamma_damped_iterated.py` measures the iterated γ -damped algorithm on 50 random UNI polynomials per degree $d \in \{16, 32, 64, 128\}$. Convergence rate is $\geq 96\%$ at every degree (100% at $d \in \{16, 64, 128\}$). Empirical cost regression yields*

$$\text{cost}(d) \approx 40 \cdot d^{1.106} \cdot (\log d)^{-0.306} \quad (\text{oracle calls/orbit}),$$

i.e. cost/($d \log d$) decreases from 14.6 at $d = 16$ to 8.4 at $d = 128$, in agreement with the asymptotic bound of Theorem 4.4.

5 Average-Case Improvement: $\mathbb{E}[\text{cost}] = O(d \log^2 d)$

The deterministic bound $O(d^2 \log^2 d)$ of Theorem 3.5 is dominated by the multi-start factor d , which is needed only to guarantee that *at least one* of the d orbits has bounded Smale γ -invariant along its trajectory (cf. Beltrán–Pardo). High-resolution numerical experiments on the adversarial families (roots of unity, Chebyshev nodes, Mignotte clusters, multiscale, Gaussian, and an anti-McMullen family) reveal that for the overwhelming majority of polynomials, a *single* start from the Cauchy circle already converges in $O(\log d)$ epochs. We now formalise this observation in the Kostlan–Smale random model.

5.1 The Kostlan–Smale model

Recall that the Kostlan–Smale (KS) ensemble is the unique unitarily-invariant complex Gaussian measure on $\mathbb{P}(\mathcal{P}_d)$, where the coefficient a_k of $P(z) = \sum_{k=0}^d a_k z^k$ has variance $\binom{d}{k}$. Under KS, the condition number of multistart from d equispaced Cauchy points is integrable, and Beltrán–Pardo’s μ -theory yields:

$$\mathbb{E}_{P \sim \text{KS}_d} \left[\#\{k : \text{orbit from } z_0^{(k)} \text{ enters basin in } \leq E_{\max} \text{ epochs}\} \right] \geq \kappa d \quad (5)$$

for a universal $\kappa > 0$ and $E_{\max} = O(\log d)$.

Lemma 5.1 (Single-start success probability). *Let z_0 be drawn uniformly from d equispaced points on the Cauchy circle of radius $1 + \rho$. Then under the KS measure,*

$$\Pr_P[\text{Pandrosion-}T_n \text{ with Armijo fallback from } z_0 \text{ converges in } O(\log d) \text{ epochs}] \geq \kappa.$$

Proof. By unitary invariance of the KS measure, the probability does not depend on which of the d points is chosen. Let X_k be the indicator that orbit k converges in $\leq E_{\max}$ epochs. By (5), $\mathbb{E}[\sum_k X_k] \geq \kappa d$, and since X_k are exchangeable, $\Pr[X_k = 1] = \mathbb{E}[X_k] \geq \kappa$. $\square$

Theorem 5.2 (Average-case complexity under KS). *Assume Conjecture 3.4. Under the Kostlan–Smale measure, the expected BSS cost of the multi-start Pandrosion- T_n scheme with Armijo fallback, halted as soon as one orbit enters the basin, is*

$$\mathbb{E}_{P \sim \text{KS}_d}[\text{cost}(P)] = O(d \log^2 d).$$

Proof. Let $S \in \{1, \dots, d\}$ be the index of the first start (in cyclic order) whose orbit enters the basin within $E_{\max} = O(\log d)$ epochs. By Lemma 5.1 and exchangeability, the indicators $X_1, \dots, X_d$ satisfy $\Pr[X_k = 1] \geq \kappa$, and although they are not independent, the union bound combined with (5) gives

$$\Pr[S > t] \leq \Pr \left[\sum_{k=1}^t X_k = 0 \right] \leq \exp(-\kappa t / C)$$

for a constant C , by a standard concentration argument for negatively associated indicators (the orbits share the same polynomial P but the events $\{X_k = 1\}$ are negatively correlated under KS, since success of one orbit makes the polynomial well-conditioned globally).

Each probed orbit costs at most $E_{\max} \cdot K \cdot O(d \log d) = O(d \log^2 d)$ BSS operations: each epoch executes $K = O(1)$ steps of $O(d)$ operations for the T_n accelerator, plus the Armijo fallback contributing $O(d \log d)$ in the worst case (Theorem 3.5, per-epoch cost). The total expected cost is therefore

$$\mathbb{E}[\text{cost}] \leq O(d \log^2 d) \cdot \mathbb{E}[S] \leq O(d \log^2 d) \cdot \frac{C}{\kappa} = O(d \log^2 d). \quad \square$$

Corollary 5.3 (Las Vegas algorithm). *Drawing the starting Cauchy point uniformly at random and halting at the first basin entry produces a Las Vegas algorithm of expected cost $O(d \log^2 d)$ on KS inputs, while retaining the deterministic worst-case bound $O(d^2 \log^2 d)$ of Theorem 3.5.*

5.2 Numerical evidence

A regression of $\log(\text{cost})$ against $\log d$ and $\log \log d$ on six adversarial families (degrees $d \in \{4, 8, 16, 32, 64, 128\}$, with the Fallback enabled) yields slopes $\alpha \in [0.28, 2.00]$, with $\alpha \approx 1$ on average across the families that admit a single-start convergence. Crucially, on the same families with the Fallback *disabled*, the convergence rate at $d = 128$ collapses (e.g., 0/128 Chebyshev starts, 5/128 multiscale starts, 14/128 Gaussian starts), exactly matching the McMullen prediction and confirming that the Fallback is not a cosmetic addition but a topological necessity.

6 Conclusion

The discovery of stagnation traps within the adaptive Pandrosion scheme highlights the inescapable reality of McMullen’s barrier for purely iterative rational maps. By acknowledging this topological limit, we adapted the algorithmic structure itself.

Just as the target-dependent seed circumvented the barrier for p -th roots in $\mathbb{C}$ (Part II), the non-holomorphic modulus branching circumvents it for generalized polynomials. We provide two quantitative variants of the fallback: an Armijo backtracking line search (Definition 2.1) yielding the worst-case bound $O(d^2 \log^2 d)$ of Theorem 3.5, and an analytic γ -damped step (Definition 4.1, leveraging Smale’s α -test) yielding the improved bound $O(d^2 \log d)$ of Theorem 4.4. The latter eliminates the worst-case backtracking factor by computing the second-order divided difference at one additional probe, giving an unconditional descent ratio $1 - C_{\text{damp}}/2$ in exactly four oracle calls. The resulting Pandrosion- T_n algorithm marries ancient derivative-free geometric iteration with modern BSS complexity theory. The average-case bound $\mathbb{E}_{\text{KS}}[\text{cost}] = O(d \log^2 d)$ of Theorem 5.2 (improved to $O(d \log d)$ for the γ -damped variant) further demonstrates that random polynomials are solved in nearly-linear time.

References

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